

Chapter Notes – Electromagnetic Spectrum (Week 7–8)

Key Concepts:

- **Definition:** Range of all electromagnetic waves arranged by frequency/wavelength.
- **Order (longest wavelength → shortest):**
Radio → Microwave → Infrared → Visible Light → Ultraviolet → X-rays → Gamma rays
- **Speed of light (c):**
 $c = 3.0 \times 10^8 \text{ m/s} = 3.0 \times 10^8 \text{ m/s}$ (in vacuum)
- **Relation:**
 $c = \lambda f$
(λ = wavelength, f = frequency)
- **Applications:**
 - Radio → Communication
 - Infrared → Heat detection
 - Visible → Human vision
 - UV → Sterilization
 - X-rays → Medical imaging
 - Gamma → Cancer treatment

Example:

If a wave has a frequency of $6 \times 10^{14} \text{ Hz}$, find its wavelength.

$$\lambda = \frac{c}{f} = \frac{3.0 \times 10^8 \text{ m/s}}{6 \times 10^{14} \text{ Hz}} = 5 \times 10^{-7} \text{ m}$$

→ This corresponds to visible light (green).